CYBERVIE

Become a Certified AI Engineering Professional with CCAEP!



Hands-on Labs | Capstone Project | Portfolio & Interview Prep

Cybervie Certified AI Engineering Professional: industry-focused LLM, RAG & agent engineering training.

- For aspiring AI engineers with Python basics
- Master LLM APIs, RAG, Agents & MCP, Fine-tuning & More
- Ship with Docker, Kubernetes, GitHub Actions & MLflow

Start Your Journey Today!

CYBERVIE

What is AI Engineering?

AI Engineering is the discipline of building real-world applications and systems on top of AI models, especially Large Language Models (LLMs).

An AI Engineer does not always train a model from scratch. Instead, an AI Engineer knows how these models work from the inside, picks the right model, gives it the right context, connects it to tools and data, makes it fast and cheap to run, measures if it is doing a good job, keeps it safe, and ships it to real users.

IN SIMPLE WORDS

Understanding the model

- Knows how models work from the inside
- Picks the right model

+

Building on top of the model

- Gives it the right context
- Connects it to tools and data

+

Running the model in production

- Makes it fast and cheap to run
- Measures if it is doing a good job
- Keeps it safe
- Ships it to real users

= AI Engineering

This is exactly what we will learn in this AI Engineering Course.

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CERTIFIED
AI ENGINEERING
PROFESSIONAL
CCAEP

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Career Map :
Roles You Can Grow Into

CCAEP prepares you for roles that build real applications on top of AI models. Start by shipping LLM features, then grow toward end-to-end ownership and system design.

STAGE 1
Start
- Junior AI Engineer
- LLM Application Developer

→

STAGE 2
Grow
- AI Engineer
- RAG Engineer
- AI Agent Developer
- LLMOps Engineer

→

STAGE 3
Lead
- Senior AI Engineer
- AI Solutions Architect
- AI Technical Lead

Role Snapshot

AI Engineer
Builds end-to-end LLM products: picks models, adds context and tools, adapts models and ships to users.

D1-D7

LLM Application Developer
Turns LLM APIs into features: prompts, structured outputs and tool-calling services.

D1 Foundations | D3 LLM Apps

RAG Engineer
Designs retrieval pipelines with vector search, hybrid ranking, citations and quality measurement.

D4 Retrieval & RAG | D3 LLM Apps

AI Agent Developer
Builds agents that plan, call tools safely and connect to systems through MCP.

D5 Agents & MCP | D3 LLM Apps

LLMOps Engineer
Containerizes, deploys and monitors AI services, and keeps cost, speed and reliability in check.

D7 Production & MLOps

AI Solutions Architect
Turns business needs into AI system designs, weighs trade-offs and explains them to teams and clients.

D8 Solution Design | D7 Production

Role titles and growth paths vary by company. This map is a guide to where CCAEP skills apply, not a guarantee of placement or salary.

Program Overview :
Why CCAEP?

CCAEP, the Cybervie Certified AI Engineering Professional, turns a 40-class bootcamp into a competency-based credential. Every class ends with a practical lab, and every domain is proven with real work: a Git portfolio, a deployed capstone and a live design defense.

40
Hands-on labs, one in every class

8
Weighted competency domains

5
Applied domain labs: e-commerce to enterprise

4
Assessment components to certify

Eight Competency Domains

Domain	Exam weight
D1 Engineering Foundations	10%
D2 LLM Internals	10%
D3 LLM Application Engineering	15%
D4 Retrieval & RAG	15%
D5 Agents & MCP	12%
D6 Models & Adaptation	13%
D7 Production & MLOps	15%
D8 Solution Design & Professional Practice	10%

The five applied domain labs are assessed in the capstone.

Tools Covered: **Python, FastAPI, Hugging Face, vector databases, MCP, Docker, Kubernetes, GitHub Actions, MLflow**

Not Included: No PyTorch module and no LangChain, LangGraph or CrewAI. Agents are built in plain Python.

Domain 1 :
Engineering Foundations

Topics Covered:

- Python Essentials for AI: Functions, Typing, JSON, HTTP Clients, Environment Management
- Async Python & FastAPI: async/await, Request Validation, REST Endpoints, Streaming Basics
- ML Fundamentals: Supervised Learning, Train/Test Splits, Features, Loss, Metrics
- ML Evaluation & Embeddings: Overfitting, Precision/Recall, Cosine Similarity

Practical Labs: **Reusable API Client with Retries, Async FastAPI Service, Baseline Classifier**

Domain 2 :
LLM Internals

Topics Covered:

- Tokenization: Tokens, BPE, Context Length, Token Budgets
- Embeddings & Vector Similarity: Semantic Representation
- Attention & QKV: Self-Attention, Masking, Multi-Head Attention
- Transformer Architecture: Decoder Blocks, Positional Information, Generation

Practical Labs: **Tokenizer Inspection, Semantic Similarity Explorer, Attention Visualization, Forward-Pass Trace**

Domain 3 :
LLM Application Engineering

Topics Covered:

- LLM API Integration: Provider APIs, Authentication, Streaming
- Prompt Engineering: System Prompts, Few-Shot Examples, Prompt Comparison
- Structured Outputs: JSON Schemas, Validation, Retries, Extraction
- Tool Calling: Function Schemas, Tool Selection, Execution Loops, Error Handling

Practical Labs: **Reusable LLM Client, Prompt Test Suite, Schema-Validated Extractor, Tool-Using FastAPI Assistant**

Domain 4 :
Retrieval & RAG

Topics Covered:

- RAG Foundations & Ingestion: Architecture, Document Parsing, Chunking, Metadata
- Vector Databases: Embedding Indexes, Similarity Search, Metadata Filters
- Hybrid Search: BM25, Dense Search, Fusion, Retrieval Metrics
- Reranking & RAG Quality: Cross-Encoders, Top-k, Citations, Failure Analysis

Practical Labs: **Baseline RAG System, Searchable Vector Collection, Lexical vs Dense vs Hybrid Benchmark, End-to-End RAG Assistant**

Domain 5 :
Agents & MCP

Topics Covered:

- Agent Fundamentals: Agent vs Workflow, Planning, Observation & Action, Limits
- Tool Orchestration: Tool Selection, Sequential Tasks, Error Recovery, Safe Actions
- MCP Fundamentals: Hosts, Clients, Servers, Tools, Resources
- MCP in Practice: Exposing Tools, Permissions, Validation, Troubleshooting

Practical Labs: **Plain-Python Agent Loop, Multi-Step Tool Assistant, MCP Client, MCP Server with an End-to-End Task**

Domain 6 :
Models & Adaptation

Topics Covered:

- Reasoning Models: Task Selection, Latency & Cost Trade-offs
- Multimodal Model APIs: Vision-Language Inputs, Image Understanding
- Hugging Face Ecosystem: Model Hub, Tokenizers, Datasets, Pipelines, Model Cards
- Fine-Tuning: Supervised Fine-Tuning, Dataset Preparation, LoRA, Held-Out Evaluation

Practical Labs: **Reasoning vs Standard Benchmark, Image Q&A API, Open-Model Evaluation, Base vs Adapted Model Comparison**

Domain 7 :
Production & MLOps

Topics Covered:

- Docker Foundations: Images, Dockerfiles, Containers, Environment Variables
- Kubernetes Foundations: Pods, Services, Deployments, Configuration, Scaling
- GitHub Actions CI/CD: Automated Tests, Build Workflows, Deployment Checks
- MLflow: Runs, Parameters, Metrics, Artifacts, Model Versions
- Production Serving: Reliability, Logging, Secrets, Health Checks
- Performance & Cost: Token Accounting, Batching, Caching, Latency
- Operational Readiness: Monitoring, Rollback, Release Checks, Incident Basics

Tools Covered: **Docker, Kubernetes, GitHub Actions, MLflow**

Practical Labs: **Containerized LLM API, Local Cluster Deployment, CI Workflow, Experiment Tracking, Cost & Latency Benchmark, Deployment Runbook**

Domain 8 :
Solution Design & Career

Topics Covered:

- AI Solution Design: Requirements, Architecture, Trade-offs, Feasibility
- Technical Communication & Interviews: Architecture Explanation, Project Storytelling
- Career Planning & Portfolio: Resume, Demo Narrative, Job Targeting

Practical Labs: **One-Page Solution Design, Design Review & Mock Interview, Portfolio-Ready Project Summary**

Applied Labs :
Domain-Based AI Labs

Topics Covered:

- E-commerce: Product Retrieval, Relevance, Catalog Metadata
- Finance: Document Extraction, Numerical Consistency, Traceable Answers
- Healthcare Administration: Privacy-Aware Handling, Human Review (Non-Diagnostic)
- Customer Support: Knowledge-Base Answers, Tool Use, Escalation
- Enterprise Knowledge: Permissions, Citations, Deployment

How You Get Certified

30%
Written Exam

Scenario-based questions weighted across all eight domains.

25%
Lab Portfolio

Lab deliverables committed to Git and reviewed at checkpoints.

35%
Capstone Application

An integrated RAG and tool-calling service plus one domain lab, deployed.

10%
Design Defense

A live architecture review plus mock interview questions.

Capstone Must Have

- A working, deployed service
- Automated tests running in GitHub Actions, in a containerized build
- Cited answers with retrieval metrics reported
- Measured token cost and latency, plus a deployment runbook

Pass with 70% overall and at least 60% in every component. Successful candidates receive a verifiable CCAEP certificate and a portfolio-ready project summary.

Start Your Journey Today!

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